

Local-plan duality for divergence measures

A candidate completion of Remark 5.13 in arXiv:2510.12424

Automated mathematics run `fa_banach.001`, lane 10

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Status. Candidate full solution, likely valid, pending expert review. The mathematical content is a formal completion of a route explicitly suggested by the source, not a claim that the cited local superposition theorem is new here.

1 Source question

Ambrosio, Ikonen, Lučić, and Pasqualetto define, for a metric measure space $(X, d, \mathfrak{m})$ and $q \in (1, \infty]$, the dual dynamic cost

$$D_q(\mu) = \inf \{ \|b\|_{\text{Der}^q(X)} : b \in \text{Der}_{\mathfrak{m}}^q(X), \mathbf{div} b = \mu \}.$$

For finite $\mathfrak{m}$ and finite signed μ , their Proposition 5.12 identifies this with a cost $B_q(\mu)$ over compact rectifiable curves. After explaining the obstruction when $\mathfrak{m}(X) = \infty$, Remark 5.13 says:

When $\mathfrak{m}$ has infinite measure, in light of the superposition principle in [13], the natural extension is to plans which are concentrated on locally rectifiable curves and which have a barycenter on $L^q(\mathfrak{m})$. We leave the details for future work.

The passage occurs on PDF page 59 of arXiv:2510.12424; see Figure 1.

Remark 5.13. If we modify the definition of $B_q(\mu)$ by removing the L^1 integrability on the barycenter, namely

$$B_q^*(\mu) := \inf \left\{ \|\text{Bar}(\pi)\|_{L^q(\mathfrak{m})} \mid \pi \in \Pi(\mu^+, \mu^-), \exists \text{Bar}(\pi) \in L^q(\mathfrak{m}) \right\} \leq B_q(\mu),$$

the stronger inequality $D_q(\mu) \leq B_q^*(\mu)$ holds, even without finiteness assumptions on $\mathfrak{m}$, with the same proof. However, when $\mathfrak{m}$ is infinite it can happen that $B_q^*(\mu) < B_q(\mu) = +\infty$. This is the case for $(X, d, \mathfrak{m})$ and μ as in Remark 5.10.

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Figure 1: Remark 5.13 in the official arXiv PDF.

2 Definitions

Throughout, $(X, d, \mathbf{m})$ is a complete separable metric space equipped with a boundedly-finite Borel measure, as in the source. Let $q \in (1, \infty]$ and let $p \in [1, \infty)$ be its conjugate exponent. Write $\mathfrak{M}(X)$ for the boundedly-finite signed Borel measures.

Let $\Gamma(X)$ be the open-ended curve space of Ambrosio–Renzi–Vitillaro [2]: its elements are increasing- reparametrization classes of maps

$$\gamma \in \text{Lip}_{\text{loc}}((0, 1); X)$$

whose length inside every bounded open subset of X is finite. At either end, such a curve has a limit in X or escapes every bounded set. The curve defines a locally normal one-current T_γ . Its boundary is

$$\partial T_\gamma = \delta_{e_1(\gamma)} - \delta_{e_0(\gamma)}, \quad (2.1)$$

where a Dirac mass is omitted when the corresponding end escapes to infinity.

Definition 1 (Local plan and barycenter). *A local plan is a positive Borel measure η on the nonconstant curves in $\Gamma(X)$ for which the following two locally finite objects exist.*

(i) *Its arclength occupation measure is*

$$\nu_\eta(A) = \int_{\Gamma(X)} \int_\gamma \mathbf{1}_A ds d\eta(\gamma).$$

(ii) *Its finite-endpoint boundary is the signed measure*

$$\beta_\eta = \int_{\Gamma(X)} \partial T_\gamma d\eta(\gamma), \quad (2.2)$$

where local absolute integrability is required: $\int \|\partial T_\gamma\|(A) d\eta(\gamma) < \infty$ for every bounded Borel set A .

We say that η has L^q barycenter if $\nu_\eta = h\mathbf{m}$ for some $h \in L^q(\mathbf{m})^+$, and then write $\text{Bar}(\eta) = h$.

The measurability requirement in this definition is intrinsic: equivalently, the displayed occupation and endpoint integrals are required to define Borel measures. On the injective curves furnished by [2], arclength measure equals $\|T_\gamma\|$, whose Borel measurability is part of their construction.

For $\mu \in \mathfrak{M}(X)$ define

$$\mathbf{B}_q^{\text{loc}}(\mu) := \inf \left\{ \|\text{Bar}(\eta)\|_{L^q(\mathbf{m})} : \eta \text{ is a local plan, } \beta_\eta = -\mu, \text{ Bar}(\eta) \in L^q(\mathbf{m}) \right\}. \quad (2.3)$$

The sign in (2.3) agrees with the source: an ordinary plan starting at μ^+ and ending at μ^- has current boundary $\mu^- - \mu^+ = -\mu$.

3 Proof intuition

The missing object is already present in the two cited papers. A derivation b produces a locally normal current by integrating $b(g)$ against the coefficient of the one-form. Bounded cutoffs reduce this correspondence to the finite-mass correspondence proved in the source, showing that the current has mass exactly $|b|\mathbf{m}$ and boundary $-\text{div}b$.

The source also proves that a norm-minimizing derivation exists and is acyclic. Therefore its local current is acyclic. The local superposition theorem [2] then decomposes it into injective, possibly unbounded curves, with no cancellation of either mass or boundary. Thus the curve occupation density is exactly $|b|$, and finite curve ends give exactly $-\mu$. This constructs an optimal local plan. In the other direction, integrating any admissible local plan produces a current, hence a derivation, whose norm is bounded by the barycenter. These are the two cost inequalities.

4 The local derivation–current correspondence

Denote by $\mathbf{N}_{1,b}^q(X; \mathbf{m})$ the locally normal one-currents T for which

$$\|T\| = h\mathbf{m} \quad \text{for some } h \in L^q(\mathbf{m})^+.$$

Lemma 2 (Localized isometric correspondence). *The formula*

$$\mathbf{t}_{\text{loc}}(b)(f, dg) = \int f b(g) d\mathbf{m}, \quad f \in \text{Lip}_{bs}(X), \quad (4.1)$$

defines a linear bijection

$$\mathbf{t}_{\text{loc}} : \text{Der}_{\mathfrak{M}}^q(X) \longrightarrow \mathbf{N}_{1,b}^q(X; \mathbf{m})$$

satisfying

$$\|\mathbf{t}_{\text{loc}}(b)\| = |b|\mathbf{m}, \quad \partial \mathbf{t}_{\text{loc}}(b) = -\mathbf{div} b. \quad (4.2)$$

Moreover, subderivations correspond to subcurrents, cycles correspond to cycles, and hence b is acyclic if and only if $\mathbf{t}_{\text{loc}}(b)$ is acyclic.

Proof. The upper mass bound and boundary identity follow directly from weak locality and integration by parts:

$$|\mathbf{t}_{\text{loc}}(b)(f, dg)| \leq \int |f| |b| \text{lip}_a(g) d\mathbf{m}, \quad \partial \mathbf{t}_{\text{loc}}(b)(g) = \int b(g) d\mathbf{m} = - \int g d\mathbf{div} b.$$

Here one localizes g by a cutoff equal to one near $\text{supp } f$, so (4.1) is meaningful with the local-current test class.

It remains to see that the mass bound is exact and that the map is onto. Let U be bounded and choose $\chi \in \text{Lip}_{bs}(X)$ with $\chi = 1$ on a neighborhood of U . Then $\chi b \in \text{Der}^1(X) \cap \text{Der}^q(X)$, while

$$\mathbf{div}(\chi b) = \chi \mathbf{div} b + b(\chi)\mathbf{m}$$

is finite. The finite-mass isometric correspondence in Theorem 4.21 of the source [1] gives $\|\mathbf{t}(\chi b)\| = |\chi b|\mathbf{m}$. On U this current agrees with $\mathbf{t}_{\text{loc}}(b)$, and locality of current mass gives

$$\|\mathbf{t}_{\text{loc}}(b)\| \llcorner U = |b|\mathbf{m} \llcorner U.$$

An exhaustion by bounded sets proves the first identity in (4.2).

Conversely, for $T \in \mathbf{N}_{1,b}^q(X; \mathbf{m})$, multiply T by bounded- support cutoffs. Each resulting finite normal current belongs to the class in Theorem 4.21 of [1], hence comes from a unique integrable derivation. On regions where two cutoffs equal one, injectivity of the finite correspondence makes these derivations agree. They therefore glue to a unique $b \in \text{Der}_{\mathfrak{M}}^q(X)$ with $T = \mathbf{t}_{\text{loc}}(b)$. The already proved mass identity yields $|b|\mathbf{m} = \|T\|$, and the boundary identity yields $\mathbf{div} b = -\partial T$.

Finally, the mass identity turns $|b - c| + |c| = |b|$ into

$$\|\mathbf{t}_{\text{loc}}(b - c)\| + \|\mathbf{t}_{\text{loc}}(c)\| = \|\mathbf{t}_{\text{loc}}(b)\|,$$

which is exactly the subcurrent relation. The boundary identity identifies zero-divergence derivations with boundaryless currents. This proves the last claim. $\square$

5 Full local-plan duality

Theorem 3 (Local-plan completion of Remark 5.13). *Let $(X, d, \mathbf{m})$ be any metric measure space in the sense of [1], let $q \in (1, \infty]$, and let p be conjugate to q . Then, for every $\mu \in \mathfrak{M}(X)$,*

$$\boxed{\mathbf{B}_q^{\text{loc}}(\mu) = \mathbf{D}_q(\mu) = \mathbf{F}_p(\mu).} \quad (5.1)$$

Whenever this value is finite, the infimum defining $\mathbf{B}_q^{\text{loc}}(\mu)$ is attained. It has a minimizer η concentrated on injective curves and satisfying the no-cancellation identities

$$\nu_\eta = |b_\mu| \mathbf{m}, \quad \int_{\Gamma(X)} \|\partial T_\gamma\| d\eta(\gamma) = |\mu|, \quad (5.2)$$

as identities of boundedly-finite measures, where b_μ is a $\mathbf{D}_q(\mu)$ -minimizer.

Proof. First let η be admissible in (2.3), and put $h = \text{Bar}(\eta)$. Integrating the curve currents gives

$$T_\eta := \int_{\Gamma(X)} T_\gamma d\eta(\gamma).$$

The occupation identity and the elementary mass estimate for a curve current give

$$\|T_\eta\| \leq \nu_\eta = h \mathbf{m}. \quad (5.3)$$

The local absolute integrability in (2.2) permits Fubini's theorem for the boundary, so

$$\partial T_\eta = \beta_\eta = -\mu.$$

Lemma 2 supplies $b_\eta \in \text{Der}_\mathfrak{M}^q(X)$ with $T_\eta = \mathbf{t}_{\text{loc}}(b_\eta)$. Equations (4.2) and (5.3) imply

$$\text{div} b_\eta = \mu, \quad |b_\eta| \leq h \quad \mathbf{m}\text{-a.e.}$$

Therefore

$$\mathbf{D}_q(\mu) \leq \|b_\eta\|_{\text{Der}^q} \leq \|h\|_{L^q(\mathbf{m})}.$$

Taking the infimum over η proves $\mathbf{D}_q(\mu) \leq \mathbf{B}_q^{\text{loc}}(\mu)$.

Conversely, suppose $\mathbf{D}_q(\mu) < \infty$. Theorem 5.9 of [1] provides a minimizer b_μ and proves that every such minimizer is acyclic. Set $T = \mathbf{t}_{\text{loc}}(b_\mu)$. By Lemma 2,

$$\|T\| = |b_\mu| \mathbf{m}, \quad \partial T = -\mu, \quad T \text{ is acyclic.} \quad (5.4)$$

The metric space X is Polish, so Theorem 1.4 of Ambrosio–Renzi–Vitillaro [2] applies to T . It yields a positive Borel measure η on $\Gamma(X)$ such that

$$T = \int T_\gamma d\eta(\gamma), \quad \|T\| = \int \|T_\gamma\| d\eta(\gamma). \quad (5.5)$$

Because T is acyclic, their theorem also says that η -almost every curve is injective and that

$$\|\partial T\| = \text{int} \|\partial T_\gamma\| d\eta(\gamma). \quad (5.6)$$

For an injective locally rectifiable curve, current mass is arclength. Combining (5.4) and (5.5) therefore gives $\nu_\eta = |b_\mu| \mathbf{m}$. Equation (5.6) gives the required local absolute integrability of the endpoint boundary, and weak integration in (5.5) gives $\beta_\eta = \partial T = -\mu$. Hence η is admissible and

$$\mathbf{B}_q^{\text{loc}}(\mu) \leq \|\text{Bar}(\eta)\|_{L^q} = \|b_\mu\|_{\text{Der}^q} = \mathbf{D}_q(\mu).$$

This proves equality and attainment. If $\mathbf{D}_q(\mu) = \infty$, the first inequality shows that no finite-cost local plan exists, so equality holds in the extended sense. Finally, $\mathbf{D}_q = \mathbf{F}_p$ is Theorem 5.9 of [1]. $\square$

6 Endpoint characterization and compatibility

For the optimal plan in Theorem 3, let η_B be its restriction to curves with two finite endpoints, let η_{BL} be the restriction to curves with only a finite left endpoint, and let η_{BR} be the restriction to curves with only a finite right endpoint. Acyclicity rules out a bi-infinite component. Remark 4.19 of [2], together with $\partial T = -\mu$, gives

$$(e_0)_\# \eta_B + (e_0)_\# \eta_{BL} = \mu^+, \quad (e_1)_\# \eta_B + (e_1)_\# \eta_{BR} = \mu^-. \quad (6.1)$$

Thus the theorem is not merely a cost identity: it characterizes the positive and negative parts of every divergence measure as finite endpoints of a no-cancellation family of local curves. When μ is finite,

$$\mu^+(X) - \mu^-(X) = \eta_{BL}(\Gamma(X)) - \eta_{BR}(\Gamma(X)), \quad (6.2)$$

so the total-mass imbalance is exactly net transport to infinity.

If $\mathbf{m}(X) < \infty$ and μ is finite, then $L^q(\mathbf{m}) \subseteq L^1(\mathbf{m})$. The optimal plan has finite total arclength, hence almost every curve has finite length and extends, by completeness, to a compact rectifiable curve. Boundary noncancellation and finiteness of μ also make the plan itself finite. Theorem 3 then reduces to Proposition 5.12 of [1]. Thus the local definition is a genuine extension rather than a competing normalization.

7 Sanity checks and edge cases

Why rays are essential. Take $X = [0, \infty)$ with Lebesgue measure, $q = \infty$, and $b(f) = f'$ almost everywhere. Then $|b| = 1$ and

$$\int_0^\infty f'(x) dx = -f(0),$$

so $\text{div} b = \delta_0$. The increasing ray from 0 to infinity has barycenter 1 and current boundary $-\delta_0$. Compact curves cannot have a single finite endpoint. The local curve space is therefore indispensable.

The source's non- L^1 example. In Remark 5.10 of [1], an interval of length n^2 carries curve weight n^{-2} . The barycenter has squared L^2 norm $\sum_n n^2 n^{-4} = \sum_n n^{-2} < \infty$, while its L^1 norm is $\sum_n n^2 n^{-2} = \infty$. This local plan consists of compact curves but is excluded from the original dynamic-plan class, exactly as expected.

The endpoint $q = \infty$. No global integrability is used. An L^∞ barycenter is integrable on bounded sets because $\mathbf{m}$ is boundedly finite, which is precisely what is needed to define the local current and all cutoff arguments.

8 Verification and novelty status

The proof was checked against the exact source statements: Theorems 4.21 and 5.9 and Remark 5.13 of [1], and Theorem 1.4 plus Remark 4.19 of [2]. Separate audits were made of the current/divergence sign, cutoff integrability, the $q = \infty$ endpoint, boundary variation, and mass escaping to infinity. The accompanying script checks the two explicit arithmetic/sign examples; it is not used as proof.

A bounded literature search on 13 August 2026 used the arXiv id and exact title of the source, the exact future-work phrase, author names, and the terms “local dynamic cost”, “locally rectifiable plan”, “divergence measure”, and “ L^q barycenter”. It checked arXiv, the authors’ publication pages, CVGMT, and the journal page for [2]. No separate paper claiming this completion was found; the journal page for [2] listed no citing papers. Novelty confidence is *moderate*: the full statement and proof were not found, but the source explicitly points to the decisive theorem and calls the remaining work “details”.

The main human-review focus should be Lemma 2, especially the cutoff/gluing passage for the inverse map, and the local absolute-integrability condition in Definition 2.1. No unproved mathematical dependency remains in the packet.

References

- [1] L. Ambrosio, T. Ikonen, D. Lučić, and E. Pasqualetto, *Metric Sobolev spaces II: dual energies and divergence measures*, arXiv:2510.12424, 2025.
- [2] L. Ambrosio, F. Renzi, and F. Vitillaro, *The superposition principle for local 1-dimensional currents*, *Nonlinear Analysis* **262** (2026), 113913; arXiv:2503.18157; doi:10.1016/j.na.2025.113913.